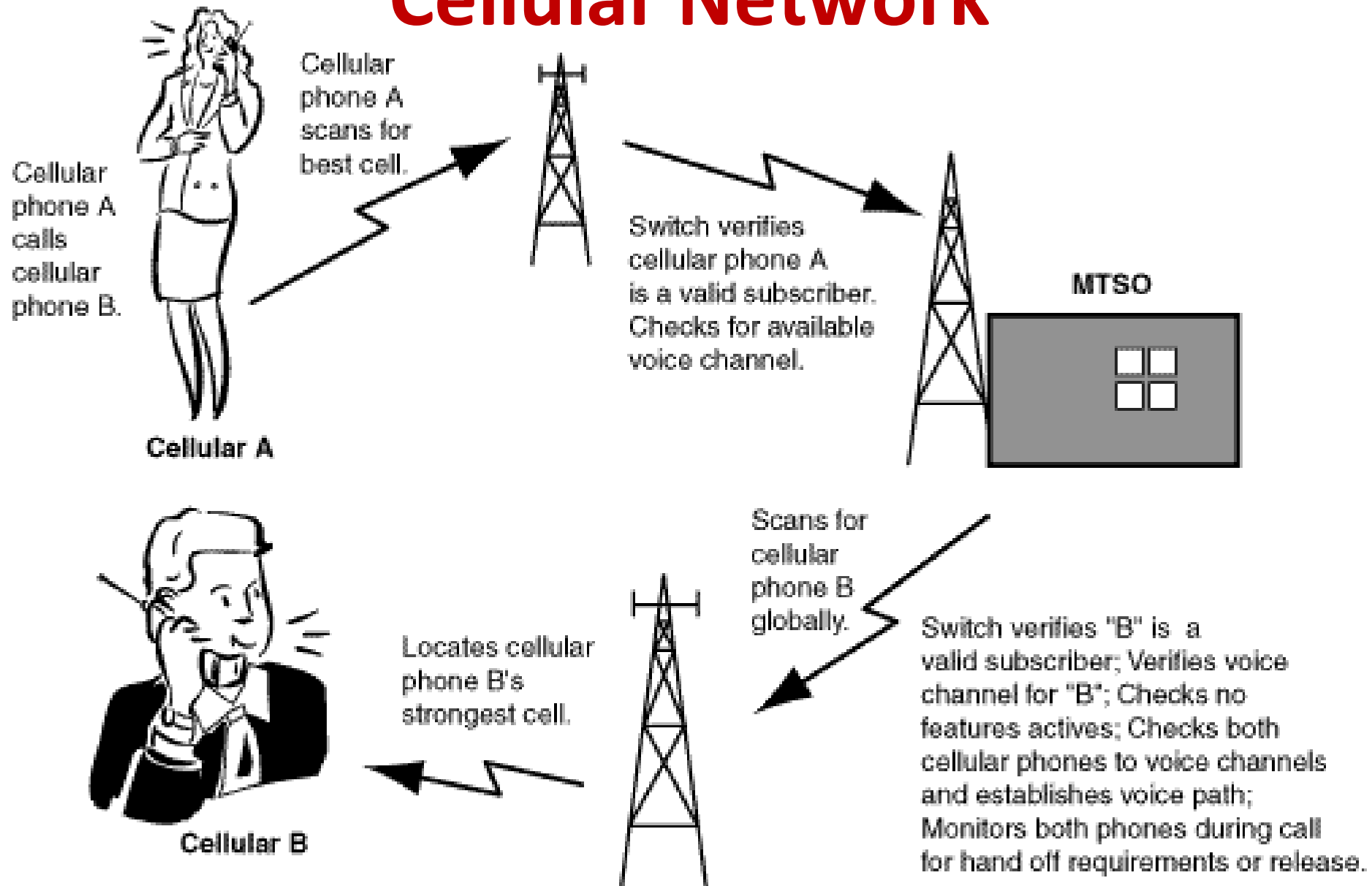
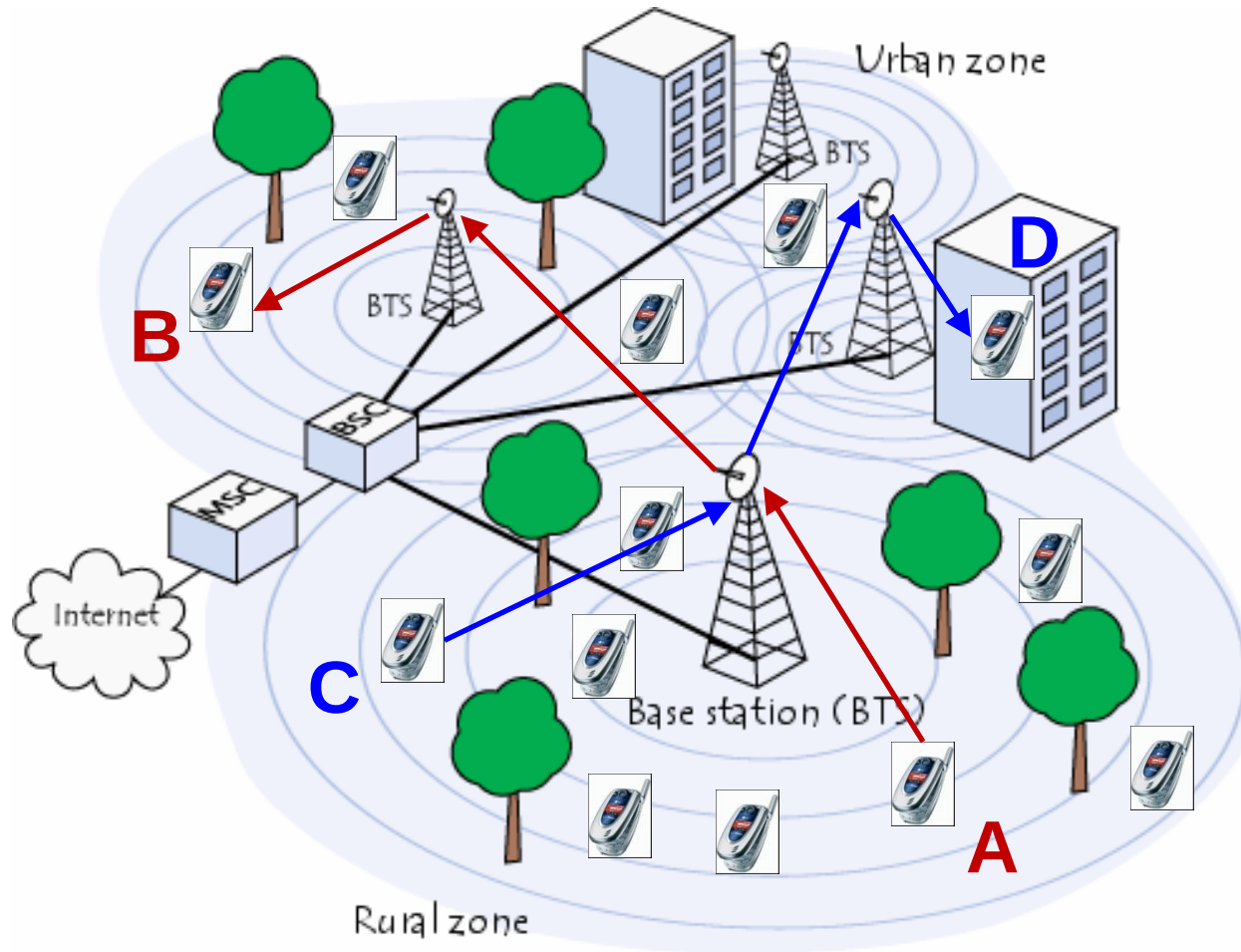


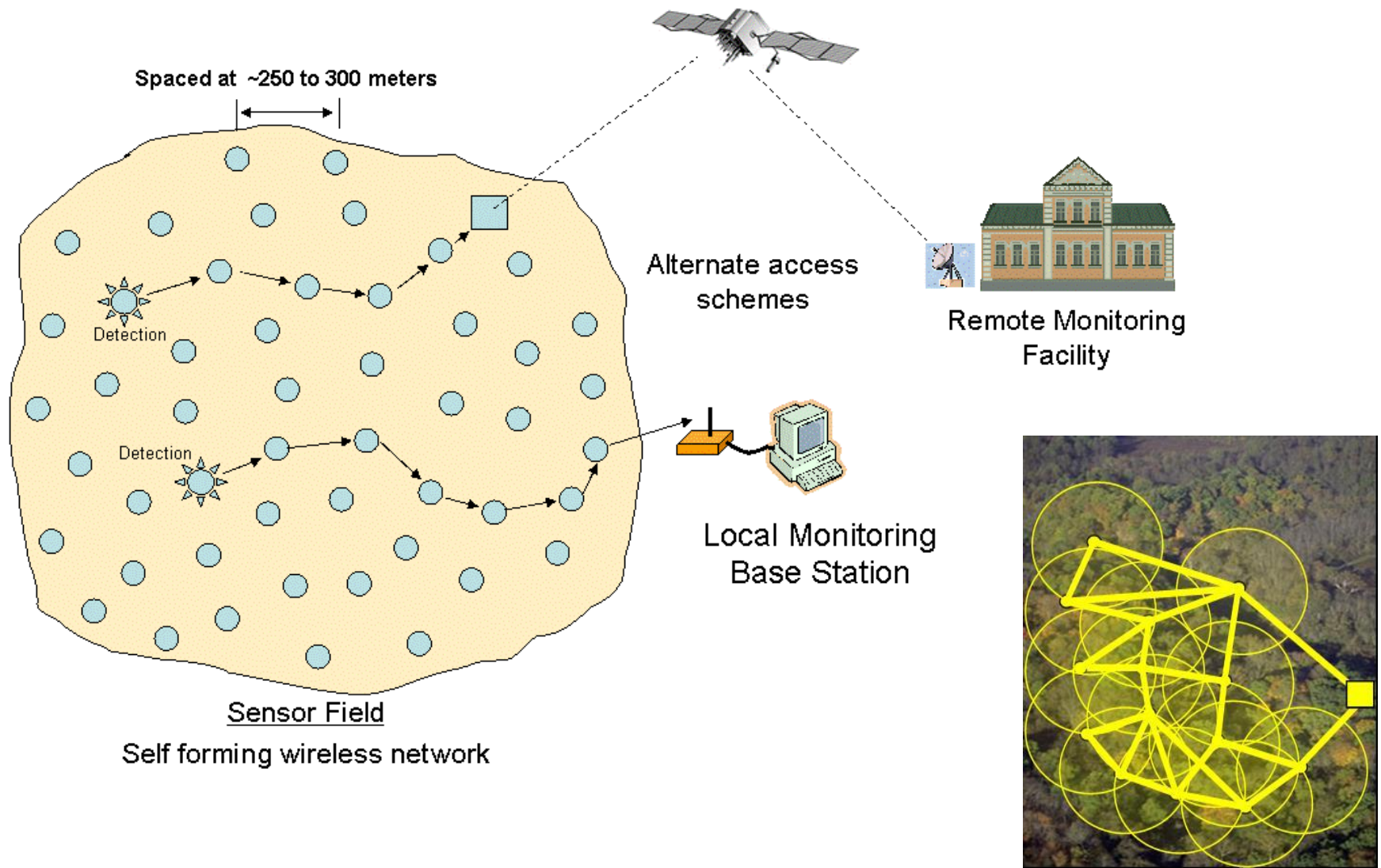
Cellular Network

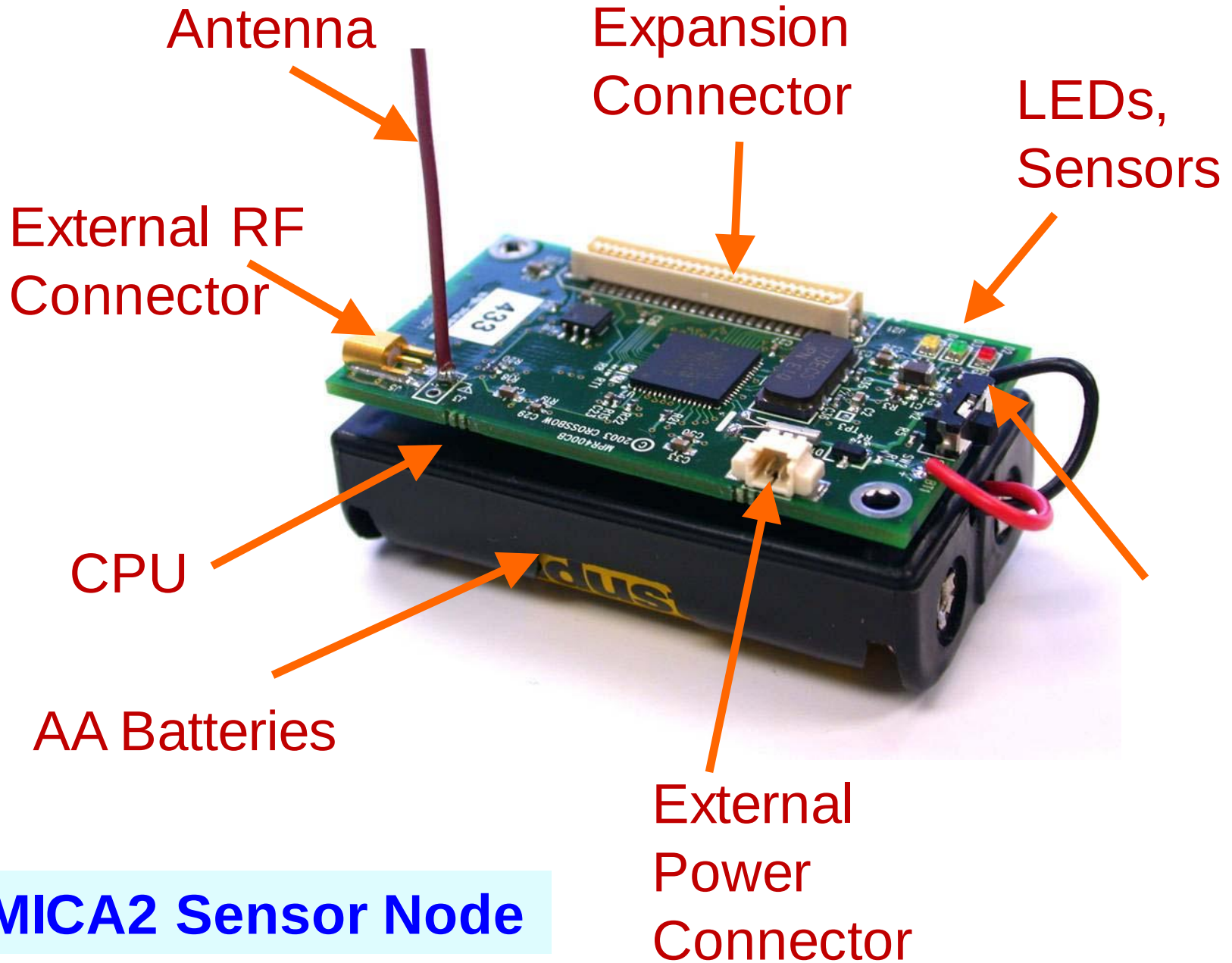


Cellular Network

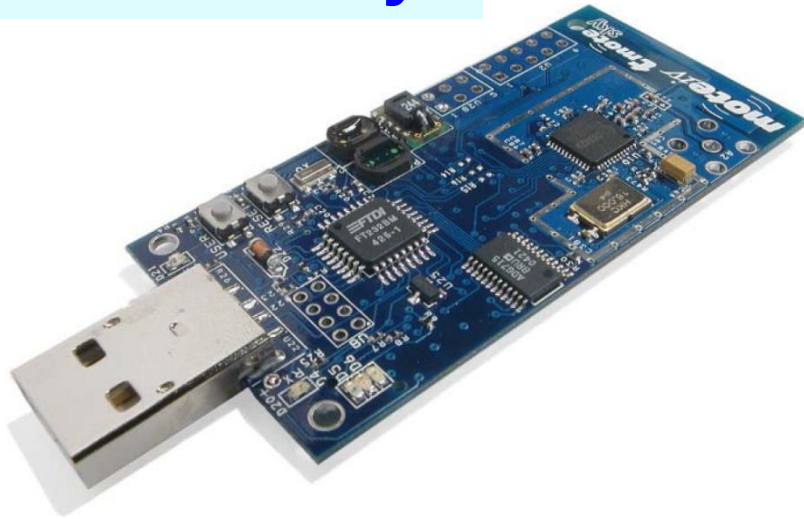


A Network without Base Station

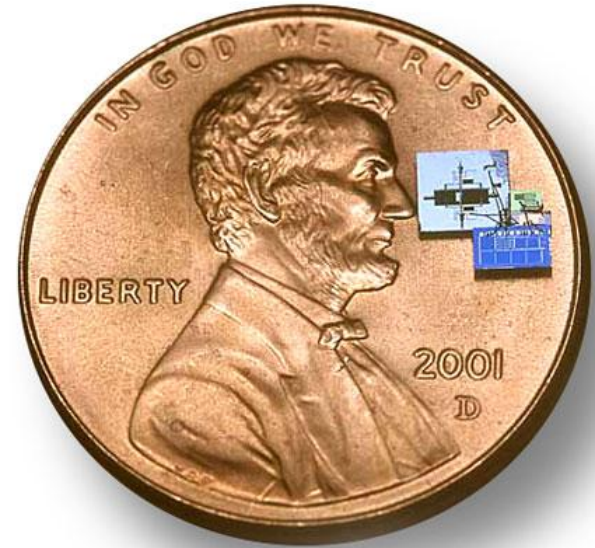




Tmote Sky



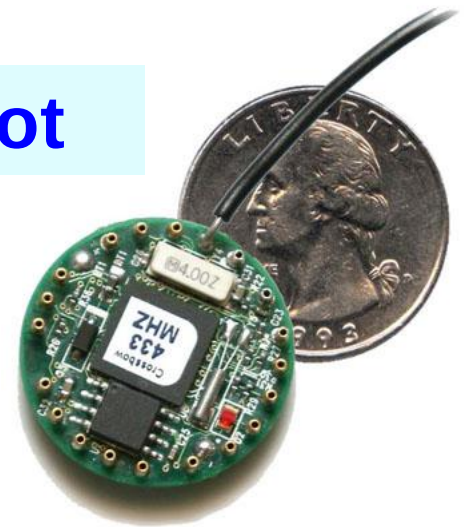
Smart Dust

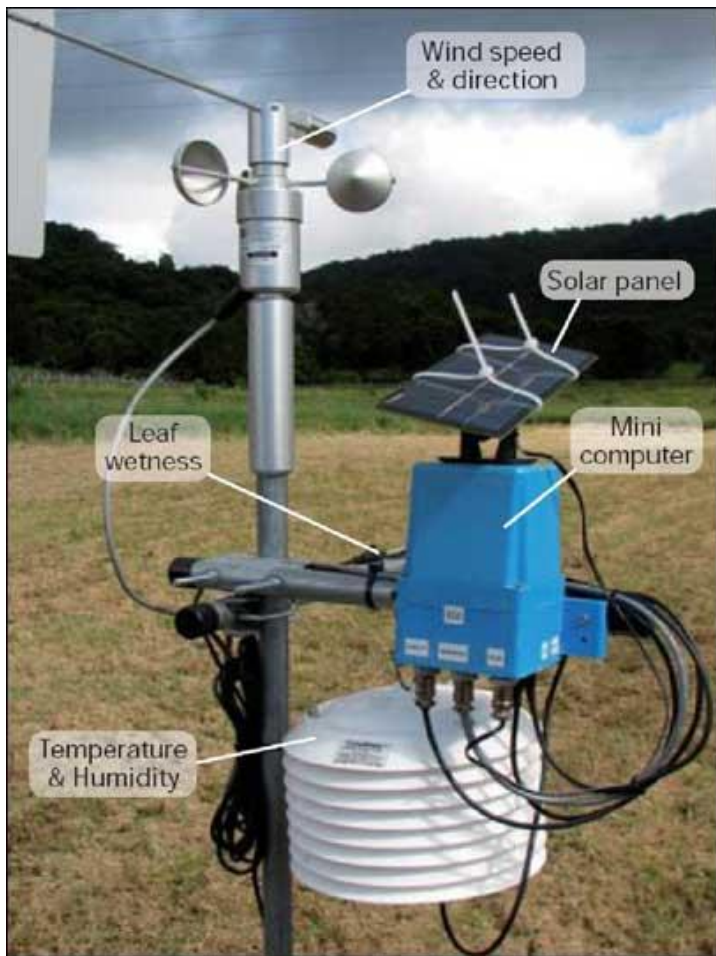


Sun SPOT



MICA Dot



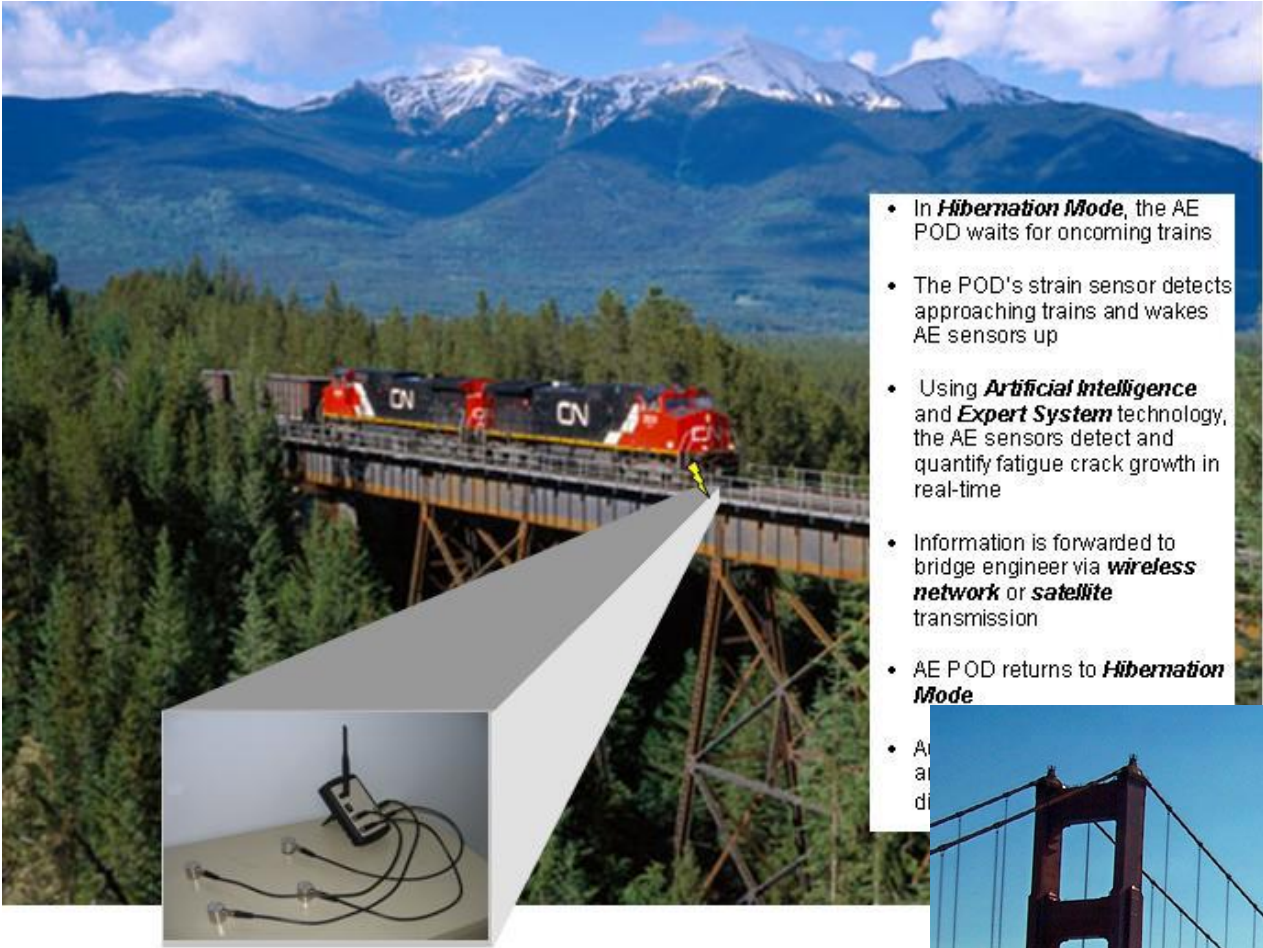


Kim Komenich / The Chronicle

**“Talking” Plants –
send text messages
when thirsty**

**Precision Agriculture –
Monitor temperature, light,
soil quality, humidity.**



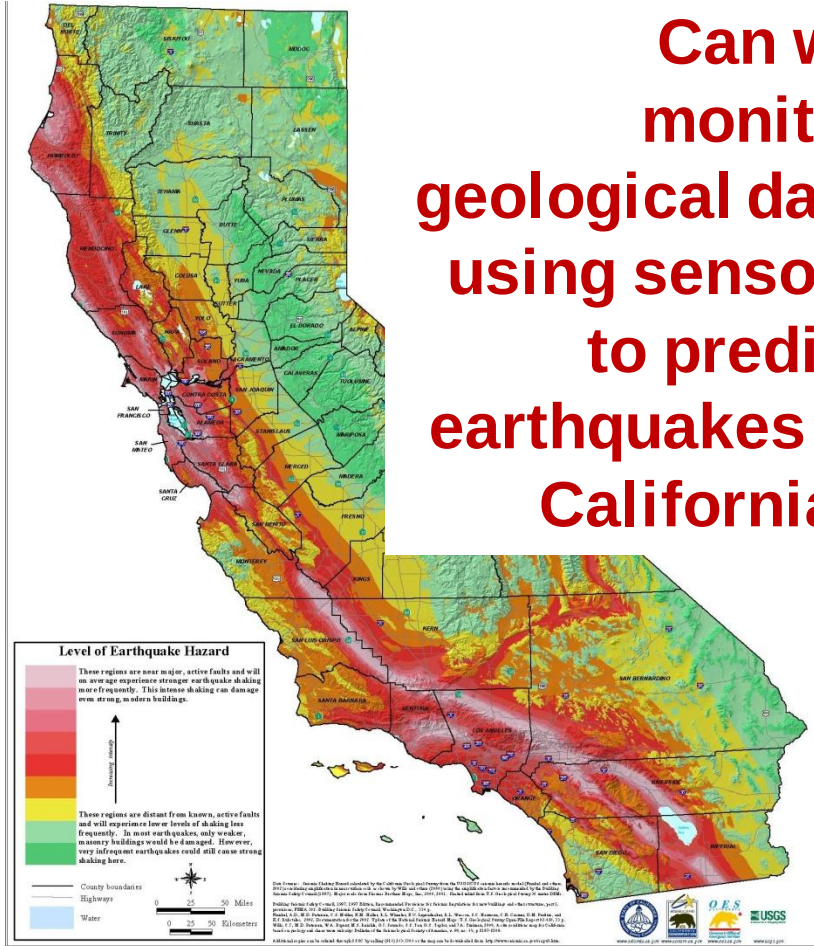


- In **Hibernation Mode**, the AE POD waits for oncoming trains
- The POD's strain sensor detects approaching trains and wakes AE sensors up
- Using **Artificial Intelligence** and **Expert System** technology, the AE sensors detect and quantify fatigue crack growth in real-time
- Information is forwarded to bridge engineer via **wireless network** or **satellite** transmission
- AE POD returns to **Hibernation Mode**
- An
a
di

Bridge monitoring for structural damages



**Can we
monitor
geological data
using sensors
to predict
earthquakes in
California?**



**Can we prevent
wildfire in California
using sensors?**